

INTRODUCTION TO Django

Django is a high-level Python web framework used to develop secure and scalable websites quickly. It follows the MVT (Model-View-Template) architecture.

Django Architecture (MVT)

Model: Handles database and defines structure of data.

View: Contains business logic and processes requests.

Template: Handles front-end (HTML pages).

Basic Django Workflow

- User sends request (URL)
- URL mapped in urls.py
- View processes request
- Data from Model (if needed)
- Template displays output
- Response sent to browser

Important Files

- models.py – Database design
- views.py – Logic handling
- urls.py – URL routing
- templates/ – HTML files
- settings.py – Project settings

Installation Commands

```
python -m pip install Django
django-admin startproject myproject
cd myproject
python manage.py startapp myapp
python manage.py runserver
```

Exp-9

PYTHON DJANGO PROJECT TO PRINT “Hello World”

AIM: To create a Python Django project to print “Hello World”.

```
django-admin startproject myproject  
cd myproject
```

PROGRAM

views.py

```
from django.http import HttpResponse
```

```
def home(request):  
    return HttpResponse("<h1>Hello Hello world !....</h1>")
```

urls.py

```
from django.urls import path
```

```
from . import views
```

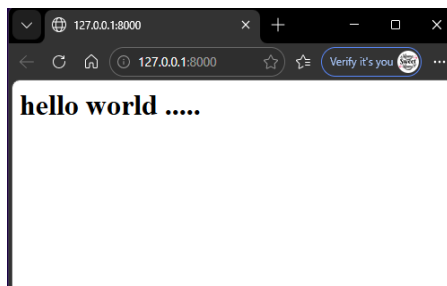
```
urlpatterns = [  
    path('admin/', admin.site.urls),  
    path(' ',views.home)  
]
```

RESULT

Program executed successfully and obtained the output.

OUTPUT

```
python manage.py runserver
```



Exp. 10

DJANGO WEB APPLICATION HOME, ABOUT AND CONTACT PAGES

AIM: To create a Django web application with three pages Home, About, and Contact Pages

```
django-admin startproject myproject
```

```
cd myproject
```

```
python manage.py startapp myapp
```

PROGRAM

1 setting.py

```
INSTALLED_APPS = [  
    .....,  
    .....,  
    'myapp',  
]
```

2 views.py

```
from django.shortcuts import render  
  
from django.http import HttpResponse  
  
def home(request):  
    return HttpResponse("Welcome to the Home Page!")  
  
def contact(request):  
    return HttpResponse("Contact us at contact@example.com")
```

3 urls.py(myproject1)

```
from django.contrib import admin  
  
from django.urls import include, path  
  
urlpatterns = [  
    path('admin/', admin.site.urls),
```

4 urls.py (myapp create new file urls.py)

```
from django.urls import path  
  
from . import views  
  
urlpatterns = [  
    path("", views.home, name='home'),
```

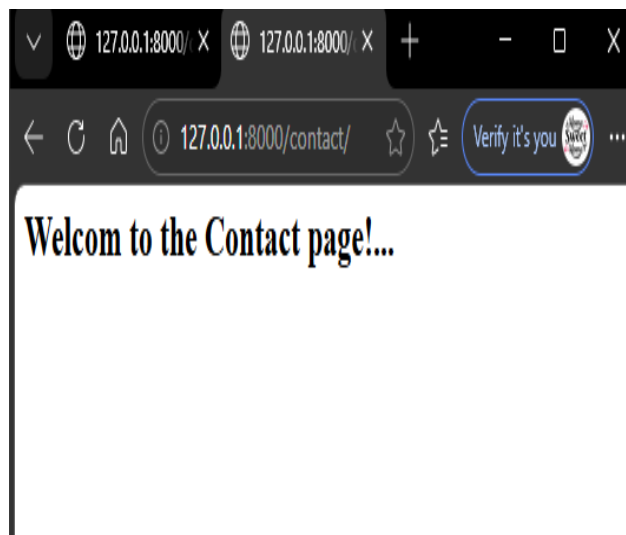
```
path("contact/", views.contact, name='contact'),  
]  
path("", include('myapp.urls')),  
]
```

RESULT

Program executed successfully and obtained the output.

OUTPUT

python manage.py runserver



Exp-11

DJANGO WEB APPLICATION USING HTML PAGE

AIM: To create a Django web application with three pages Home, About, and Contact using html page.

1-Create Project & App

```
django-admin startproject mysite .  
python manage.py startapp main
```

PROGRAM

2- Register App- mysite/settings.py

```
INSTALLED_APPS = [  
'django.contrib.admin',  
'django.contrib.auth',  
'django.contrib.contenttypes',  
'django.contrib.sessions',  
'django.contrib.messages',  
'django.contrib.staticfiles',  
'main',  
]
```

3-Create Views- main/views.py

```
from django.shortcuts import render  
  
def home(request):  
    return render(request, 'home.html')  
  
def about(request):  
    return render(request, 'about.html')  
  
def contact(request):  
    return render(request, 'contact.html')
```

4 URL Configuration-main/urls.py

```
from django.urls import path  
from . import views  
urlpatterns = [  
    path("", views.home, name='home'),  
    path('about/', views.about, name='about'),
```

```
    path('contact/', views.contact, name='contact'),  
]
```

4-1 mysite/urls.py

```
from django.contrib import admin  
from django.urls import path, include  
urlpatterns = [  
    path('admin/', admin.site.urls),  
    path("", include('main.urls')),  
]
```

5-Create Templates-main/templates/

Home.html

```
<!DOCTYPE html>  
  
<html>  
  
<head>  
    <title>Home</title>  
</head>  
  
<body>  
    <h1>Welcome to Home Page</h1>  
    <a href="/about/">About</a> |  
    <a href="/contact/">Contact</a>  
</body>  
</html>
```

about.html

```
<!DOCTYPE html>  
  
<html>  
  
<head>  
    <title>About</title>  
</head>  
  
<body>  
    <h1>About Us</h1>
```

```
<p>This is the about page.</p>
<a href="/">Home</a>
</body>
</html>

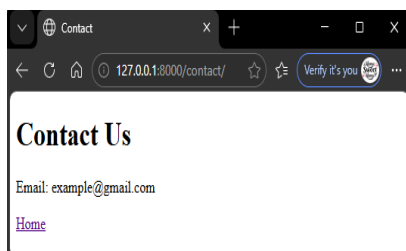
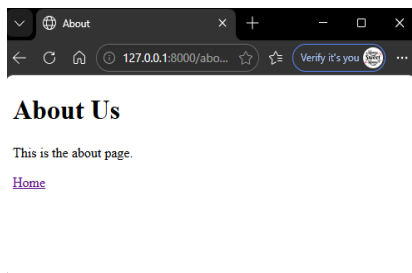
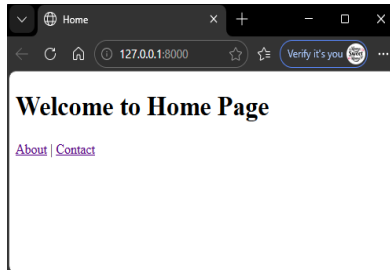
contact.html
<!DOCTYPE html>
<html>
<head>
  <title>Contact</title>
</head>
<body>
  <h1>Contact Us</h1>
  <p>Email: example@gmail.com</p>
  <a href="/">Home</a>
</body>
</html>
```

RESULT

Program executed successfully and obtained the output.

OUTPUT

python manage.py runserver



Exp-12

DJANGO PROGRAM TO DISPLAY A LIST ON A WEB PAGE

AIM: To create a Django program to display a list of items('Apple', 'Banana', 'Mango', 'Orange') on a webpage.

```
django-admin startproject myproject .  
python manage.py startapp listapp
```

PROGRAM

1-settings.py

```
INSTALLED_APPS = [  
'listapp',  
]
```

2-listapp/views.py:

```
from django.shortcuts import render  
  
def item_list(request):  
    items = ['Apple', 'Banana', 'Mango', 'Orange']  
    return render(request, 'listapp/item_list.html', {'items': items})
```

3-listapp/urls.py:

```
from django.urls import path  
  
from . import views  
  
urlpatterns = [  
    path("", views.item_list),  
]
```

4-main urls.py:

```
from django.urls import path, include  
  
urlpatterns = [  
    path("", include('listapp.urls')),  
]
```

5-listapp/templates/listapp/ item_list.html

```
<!DOCTYPE html>  
  
<html>
```

```
<head>
  <title>Simple List</title>
</head>
<body>
  <h1>Fruit List</h1>
  <ul>
    {% for item in items %}
      <li>{{ item }}</li>
    {% endfor %}
  </ul>
</body>
</html>
```

RESULT

Program executed successfully and obtained the output.

OUTPUT

python manage.py runserver

